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A Regression-Based Adjusted Plus-Minus Statistic for NHL Players

Brian Macdonald

Abstract

The goal of this paper is to develop an adjusted plus-minus statistic for NHL players that is independent of both teammates and opponents. We use data from the shift reports on NHL.com in a weighted least squares regression to estimate an NHL player's effect on his team's success in scoring and preventing goals at even strength. Both offensive and defensive components of adjusted plus-minus are given, estimates in terms of goals per 60 minutes and goals per season are given, and estimates for forwards and defensemen are given.

KEYWORDS: NHL, plus-minus, adjusted plus-minus

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1 Introduction

Hockey analysts have developed several metrics that attempt to quantify an NHL player's contribution to his team. Tom Awad's Goals Versus Threshold in Awad (2009), Jim Corsi's Corsi rating as described in Boersma (2007), Gabriel Desjardins' Behindthenet Rating, along with his on-ice/off-ice, strength of opponents, and strength of linemates statistics in Desjardins (2010), Iian Fyffe's Point Allocation in Fyffe (2002), Ken Krzywicki's shot quality, as presented in Krzywicki (2005) and updated in Krzywicki (2009), Alan Ryder's Player Contribution in Ryder (2004), and Timo Seppa's Even-Strength Total Rating in Seppa (2009) are a few examples. In this paper, we propose a new metric, adjusted plus-minus (*APM*), that attempts to estimate a player's contribution to his team in even-strength (5-on-5, 4-on-4, and 3-on-3) situations, independent of that player's teammates and opponents, and in the units of goals per season. *APM* can also be expressed in terms of goals per 60 minutes (*APM*/60). We find both an adjusted offensive plus-minus component (*OPM*) and an adjusted defensive plus-minus component (*DPM*), which estimate the offensive and defensive production of a player at even-strength, independent of teammates and opponents, and in the units of goals per season.

Inspired by the work in basketball by Rosenbaum (2004), Ilardi and Barzilai (2008), and Witus (2008), we use weighted multiple linear regression models to estimate *OPM* per 60 minutes (*OPM*/60) and *DPM* per 60 minutes (*DPM*/60). The estimates are a measure of the offensive and defensive production of a player in the units of goals per 60 minutes. These statistics, along with average minutes played per season, give us *OPM* and *DPM*. Adding *OPM*/60 and *DPM*/60 gives *APM*/60, and adding *OPM* and *DPM* gives *APM*. We emphasize that we consider only even-strength situations.

The main benefit of the weighted linear regression model is that the resulting adjusted plus-minus statistics for each player should in theory be independent of that player's teammates and opponents. The traditional plus-minus statistic in hockey is highly dependent on a player's teammates and opponents, and the use of the regression removes this dependence. One drawback of our model is statistical noise. In order to improve the estimates and reduce the standard errors in the estimates, we use data from three NHL seasons.

1.1 Example of the results

Before we describe the models in detail, we give the reader an example of the results from the Rosenbaum model. The typical NHL fan has some idea of who the best offensive players in the league are, so we give the top 10 players in average

OPM during the 2007-08, 2008-09, and 2009-10 regular seasons, sorted by *OPM*, in Table 1.1.1. Note that *Rk* is the rank of that player in terms of *OPM*, *Pos* is

Table 1.1.1: Top 10 Skaters in *OPM*

Rk	Player	Pos	OPM	DPM	APM	AErr	GF	Mins
1	Pavel Datsyuk	C	30.0	11.5	41.5	13.8	67	1200
2	Alex Ovechkin	LW	29.7	0.1	29.8	15.2	78	1272
3	Henrik Sedin	C	29.6	-12.3	17.3	18.1	66	1178
4	Sidney Crosby	C	28.1	-3.1	25.0	10.3	64	1063
5	Evgeni Malkin	C	25.2	-5.4	19.8	10.9	66	1172
6	Zach Parise	LW	24.1	3.8	27.9	12.9	57	1176
7	Eric Staal	C	24.0	0.5	24.5	12.1	58	1168
8	Joe Thornton	C	23.0	5.6	28.5	12.6	64	1228
9	Ilya Kovalchuk	LW	22.3	-7.0	15.3	11.2	59	1198
10	Marian Gaborik	RW	21.6	7.6	29.2	8.9	48	865

the player's position, *AErr* is the standard error in the *OPM* estimates, *GF* are the goals per season that a player's team scored while he was the ice at even-strength, and *Mins* is the number of minutes that the player played on average during the 2007-08, 2008-09, and 2009-10 regular seasons. The 10 players in this list are arguably the best offensive players in the game. Ovechkin, Crosby, Datsyuk, and Malkin, perhaps the league's most recognizable superstars, make the top 5 along with Henrik Sedin, who led the NHL in even-strength points during the 2009-2010 season with 83 points, which was 10 more points than the next leading scorer.

We highlight two interesting numbers in this list. First, note that Pavel Datsyuk, who is regarded by many as the best two-way player in the game, has the highest defensive rating among these top offensive players. Datsyuk's excellent two-way play gives him the highest *APM* estimate in the league. We give the list of top 10 forwards and defensemen in *APM*, and discuss several other top 10 lists for *OPM/60*, *OPM*, *DPM/60*, *DPM*, *APM/60*, and *APM*, in Section 3. Second, note that Henrik Sedin has a much higher *AErr* than the other players in the list. This increased error is likely due to the fact that Henrik plays most of his minutes with his brother Daniel, and the model has a difficult time separating the contributions of the twin brothers. The Sedin twins provide us with a great example to use when analyzing the errors, and we discuss the Sedins and the errors in more detail in Section 4.6.

1.2 Complete Results

A .csv file containing the complete results from the Rosenbaum model can be obtained by contacting the author. An interested reader may prefer to open these results in a spreadsheet program and filter by position or sort by a particular statistic. Also, the .csv file contains more columns than the list given in Table 1.1.1. For example, the file includes the three most frequent linemates of each player along with the percentage of minutes played with each of those linemates during the 2007-08, 2008-09, and 2009-10 NHL regular seasons. An example of these additional columns is given in Table 1.2.1. Notice that, as suggested in the table, Henrik

Table 1.2.1: Example of Linemate Details

Player	Pos	Mins	Teammate.1	min1	Teammate.2	min2	Teammate.3	min3
Henrik Sedin	C	1178	D.Sedin	83%	A.Edler	35%	A.Burrows	34%
Daniel Sedin	LW	1066	H.Sedin	92%	A.Edler	35%	A.Burrows	33%

Sedin played 83% of his minutes with brother Daniel, and Daniel played 92% of his minutes with Henrik.

Finally, the file includes columns for the goals a player's team scored (GF), the goals a player's team allowed (GA), and the net goals a player's team scored (NG), while he was on the ice at even-strength. These statistics are in terms of average goals per season during the 2007-08, 2008-09, and 2009-10 NHL regular seasons. We also give $GF/60$, $GA/60$, and $NG/60$, which are GF , GA , and NG in terms of goals per 60 minutes. An example of this information is given in Table 1.2.2. These raw statistics, along with the linemate information, will be helpful in

Table 1.2.2: Example of GF, GA, and NG statistics

Player	Pos	GF60	GA60	NG60	GF	GA	NG
Pavel Datsyuk	C	3.37	1.88	1.48	67	38	29
Sidney Crosby	C	3.61	2.57	1.03	64	46	18

the analysis of the results of our model.

The rest of this paper is organized as follows. In Section 2, we describe the two models we use to compute OPM , DPM , and APM . In Section 3, we summarize and discuss the results of these models by giving various top 10 lists, indicating the

best forwards and defensemen according to *OPM*, *DPM*, and *APM*, as well as their corresponding per 60 minute statistics. Section 4 contains a discussion of the model. We summarize and discuss the advantages (4.1) and disadvantages (4.2) of these statistics. Next, we give more details about the formation of the model, including the selection of the variables (4.3), and selection of the observations (4.4). Also, we discuss our assumptions (4.5) as well as the standard errors in the estimates (4.6). We finish with ideas for future work and some conclusions (4.7).

2 Two weighted least-squares models

We now define our variables and state our models. In each model, we use players who have played a minimum of 3000 shifts over the course of the 2007-08, 2008-09, and 2009-10 regular seasons (see Section 4 for a discussion). We define a shift to be a period of time during the game when no substitutions are made. The observations in each model are weighted by the duration of that observation in seconds. We note that, unlike in Ilardi and Barzilai (2008) and Rosenbaum (2004), we do not weight observations from more recent seasons more heavily. In Ilardi and Barzilai (2008) and Rosenbaum (2004), the authors were estimating player performance in the most recent season of their study. Here, we choose to estimate player performance over a three year period and have weighted all three seasons equally. Also, we have chosen to estimate regular season performance and include only regular season data in our models.

2.1 Ilardi-Barzilai-type model

Inspired by Ilardi and Barzilai (2008), we use the following linear model:

$$y = \beta_0 + \beta_1 X_1 + \cdots + \beta_J X_J + \delta_1 D_1 + \cdots + \delta_J D_J, \quad (1)$$

where J is the number of skaters in the league. The variables in the model are defined as follows:

$$\begin{aligned} y &= \text{goals per 60 minutes during an observation} \\ X_j &= \begin{cases} 1, & \text{skater } j \text{ is on offense during the observation;} \\ 0, & \text{skater } j \text{ is not playing or is on defense during the observation;} \end{cases} \\ D_j &= \begin{cases} 1, & \text{skater } j \text{ is on defense during the observation;} \\ 0, & \text{skater } j \text{ is not playing or is on offense during the observation;} \end{cases} \end{aligned} \quad (2)$$

where $1 \leq j \leq J$. Note that by “skater” we mean a forward or a defensemen, but not a goalie. We have not included goalies in our model. See Section 4.5.1 for a discussion.

Note that for the observations in this model, each shift is split into two lines of data: one line corresponding to the goals per 60 minutes scored by home team, and one line corresponding to the goals per 60 minutes scored by the away team. Both of these observations are weighted by the duration of the shift. By the expression “on offense” in (2), we mean that the player is on the ice for the home team for an observation corresponding to the goals per 60 minutes scored by the home team, or on the ice for the away team for an observation corresponding to the goals per 60 minutes scored by the away team. The phrase “on defense” has an similar meaning.

This model can also be expressed as a system of linear equations in block matrix form in the following way:

$$\underbrace{\begin{bmatrix} y_h \\ y_a \end{bmatrix}}_{2M \times 1} = \underbrace{\begin{bmatrix} X_h & D_a \\ X_a & D_h \end{bmatrix}}_{2M \times 2J} \underbrace{\begin{bmatrix} \beta \\ \delta \end{bmatrix}}_{2J \times 1} + \beta_0 \mathbb{1}_{2M \times 1}, \quad (3)$$

where

y_h, y_a are $M \times 1$ matrices containing the goals per 60 minutes of the home and away teams, respectively, during each shift;

X_h, D_a are $M \times J$ matrices with a one in column j if skater j is on the ice for the home or away team, respectively, during an observation corresponding to y_h , and with a zero in column j otherwise;

X_a, D_h are $M \times J$ matrices with a one in column j if skater j is on the ice for the away or home team, respectively, during an observation corresponding to y_a , and with a zero in column j otherwise;

β, δ are $[\beta_1, \dots, \beta_J]^T$, and $[\delta_1, \dots, \delta_J]^T$, respectively, and

$\mathbb{1}_{2M \times 1}$ is a matrix of ones for the intercept term.

For example, suppose that in the first shift of the model, skaters 1-5 are on the ice for the home team, and skaters 6-10 are on the ice for the away team. For this shift we have two lines of data, one for goals per 60 minutes scored by the home team, and the second for goals per 60 minutes scored by the away team. The first row of the four block matrices in equation (3) is given below.

$$\begin{aligned} X_h &= [1 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ \cdots \ 0], D_a = [0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 1 \ 1 \ 0 \ \cdots \ 0], \\ X_a &= [0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 1 \ 1 \ 0 \ \cdots \ 0], D_h = [1 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ \cdots \ 0], \end{aligned}$$

Note that since the home players on the ice during observations corresponding to y_h are the same as the home players on the ice during observations corresponding to y_a , we have that $X_h = D_h$. Likewise, we have that $D_a = X_a$, and equation (3) can be written as

$$\begin{bmatrix} y_h \\ y_a \end{bmatrix} = \begin{bmatrix} X_h & D_a \\ D_a & X_h \end{bmatrix} \begin{bmatrix} \beta \\ \delta \end{bmatrix} + \beta_0 \mathbb{1}_{2M \times 1}. \quad (4)$$

The coefficients have the following interpretation:

$$\begin{aligned} \beta_j &= \text{goals per 60 minutes contributed by skater } j \text{ on offense,} \\ -\delta_j &= \text{goals per 60 minutes contributed by skater } j \text{ on defense,} \\ \beta_0 &= \text{intercept,} \end{aligned} \quad (5)$$

The least-squares regression coefficient $\hat{\beta}_1$, for example, gives an estimate, in goals per 60 minutes, of how y changes when Skater 1 is on the ice on offense ($X_1 = 1$) versus when Skater 1 is not on the ice on offense ($X_1 = 0$), independent of all other players on the ice. The least-squares regression coefficients $\hat{\beta}_j$ and $-\hat{\delta}_j$, are estimates of $OPM/60$ for Skater j and $DPM/60$ for Skater j , respectively. They are playing-time-independent rate statistics, measuring the offensive and defensive value of a player in goals per 60 minutes.

Notice the negative sign in front of δ_j in (5). Note that a negative value for one of these coefficients corresponds to a positive contribution. For example, if Skater 1 has a defensive coefficient of $\hat{\delta}_1 = -0.8$, he prevents 0.8 goals per 60 minutes when he is on defense. We could have chosen to define a skater's $DPM/60$ to be $+\hat{\delta}_j$, in which case negative values for $DPM/60$ would be good. Instead, we prefer that positive contributions be represented by a positive number, so we define $DPM/60 = -\hat{\delta}_j$ for skaters. For Skater 1's $DPM/60$ in our example, we have

$$DPM/60 = -\hat{\delta}_1 = -(-0.8) = +0.8,$$

which means that Player 1 has a positive contribution of +0.8 goals per 60 minutes on defense.

A player's $APM/60$ can be computed as follows:

$$APM/60 = OPM/60 + DPM/60. \quad (6)$$

We estimate $AErr$, the standard errors for the $APM/60$ estimates, using

$$AErr = \sqrt{\text{Var}(APM/60)} = \sqrt{(OErr)^2 + (DErr)^2 + 2r_{OD} OErr DErr},$$

where $r_{OD} \approx -0.116$, and where $OErr$ and $DErr$ are the standard errors in the $OPM/60$ and $DPM/60$ estimates, respectively.

Finally, we note that the data used for the model were obtained from the shift charts published in NHL.com (2010) for games played in the 2007-08, 2008-09, and 2009-10 regular seasons. See Section 4.4 for more about the data used and to see how it was selected.

2.2 Calculating OPM , DPM , and APM

A player's contribution in terms of goals over an entire season is useful as well, and may be preferred by some NHL fans and analysts. We use the regression coefficients and minutes played to give playing-time-dependent counting statistic versions of the rate statistics from the regression model. These counting statistics are OPM , DPM , and APM , and they measure the offensive, defensive, and total value of a player, in goals per season. To get a skater's OPM , for example, we multiply a skater's offensive contribution per minute by the average number of minutes that the skater played per season from 2007-2010. The value for DPM is found likewise, and APM for a player is the sum of his OPM and his DPM . Let

$$Min_j = \text{minutes per season for skater } j.$$

Then, we can calculate OPM , DPM , and APM for skaters as follows:

$$\begin{aligned} OPM_j &= \hat{\beta}_j Min_j / 60, \\ DPM_j &= -\hat{\delta}_j Min_j / 60, \\ APM_j &= OPM_j + DPM_j. \end{aligned} \tag{7}$$

2.3 Rosenbaum-type model

We now describe a second linear model, this one inspired by Rosenbaum (2004):

$$y_{net} = \eta_0 + \eta_1 N_1 + \cdots + \eta_J N_J. \tag{8}$$

The variables in the model are defined as follows:

y_{net} = net goals per 60 minutes for the home team during an observation

$$N_j = \begin{cases} 1, & \text{player } j \text{ is on the home team during the observation;} \\ -1, & \text{player } j \text{ is on the away team during the observation;} \\ 0, & \text{player } j \text{ is not playing during the observation.} \end{cases}$$

The model can also be expressed as a linear system in block matrix form as

$$y_{net} = N \eta + \eta_0 \mathbb{1}_{M \times 1}, \tag{9}$$

where

y_{net} is an $M \times 1$ matrix of net goals per 60 minutes scored by the home team during a shift;

N is an $M \times J$ matrix, with a one in column j if player j is on the home team and on the ice during an observation, a minus one if player j is on the away team and on the ice, and a 0 if the player is not on the ice;

η is $[\eta_1, \dots, \eta_J]^T$.

The coefficients in the model have the following interpretation:

η_j = net goals per 60 minutes contributed by player j ,
 η_0 = intercept.

The least-squares regression coefficients $\hat{\eta}_1, \dots, \hat{\eta}_J$ of this model are estimates of each player's $APM/60$. By “net goals” we mean Goals For (GF) minus Goals Against (GA). Also, note that by “player” we mean forward or defensemen. The data used for these variables were also obtained from the shift charts published on NHL.com for games played in the 2007-08, 2008-09, and 2009-10 regular seasons. In this model, unlike the Ilardi-Barzilai model, each observation in this model is simply one shift. We do not split each shift into two lines of data.

In order to separate offense and defense, we follow Rosenbaum and Witus (2008) and form a second model:

$$y_{tot} = \tau_0 + \tau_1 T_1 + \dots + \tau_J T_J. \quad (10)$$

The variables in the model are defined as follows:

y_{tot} = total goals per 60 minutes scored by both teams during an observation

$T_j = \begin{cases} 1, & \text{player } j \text{ is on the ice (home or away) during the observation;} \\ 0, & \text{player } j \text{ is not on the ice during the observation.} \end{cases}$

The model can also be expressed as a linear system in block matrix form as

$$y_{tot} = T \tau + \tau_0 \mathbf{1}_{M \times 1}, \quad (11)$$

where

y_{tot} is an $M \times 1$ matrix of total goals per 60 minutes scored by both teams during a shift;

T is an $M \times J$ matrix, with a one in column j if player j is on the ice for the home or away team during an observation, and a 0 if the player is not on the ice;

τ is $[\tau_1, \dots, \tau_J]^T$.

The coefficients in the model have the following interpretation:

τ_j = total goals per 60 minutes contributed by skater j ,
 τ_0 = intercept.

By total goals, we mean $GF + GA$. Recall that the least-squares coefficients $\hat{\eta}_j$ for (8) were estimates of each player's $APM/60$, or net goals contributed per 60 minutes. Likewise, the least-squares coefficients $\hat{\tau}_j$ for (10) are estimates of each player's $TPM/60$, or total goals contributed per 60 minutes.

We know from before that

$$APM/60 = OPM/60 + DPM/60, \quad (12)$$

and we also have that

$$TPM/60 = OPM/60 - DPM/60. \quad (13)$$

Using equations (12) and (13), if we add a player's $TPM/60$ and $APM/60$, and divide by 2, the result is that player's $OPM/60$:

$$\frac{1}{2}(APM/60 + TPM/60) = OPM/60.$$

Likewise,

$$\frac{1}{2}(APM/60 - TPM/60) = DPM/60.$$

Using playing time, we can convert $OPM/60$, $DPM/60$, and $APM/60$ to OPM , DPM , and APM as we did with our first model in Section 2.1.

3 Summary of Results

In this section we will summarize the results of the Rosenbaum model by giving various top 10 lists, indicating the best offensive, defensive, and overall players in the league according to the estimates found in the model.

3.1 $OPM/60$

Recall that $OPM/60$ is a measure of the offensive contribution of a player at even-strength in terms of goals per 60 minutes of playing time. The list of top players in $OPM/60$ is given in Table 3.1.1. The players in this list are regarded by

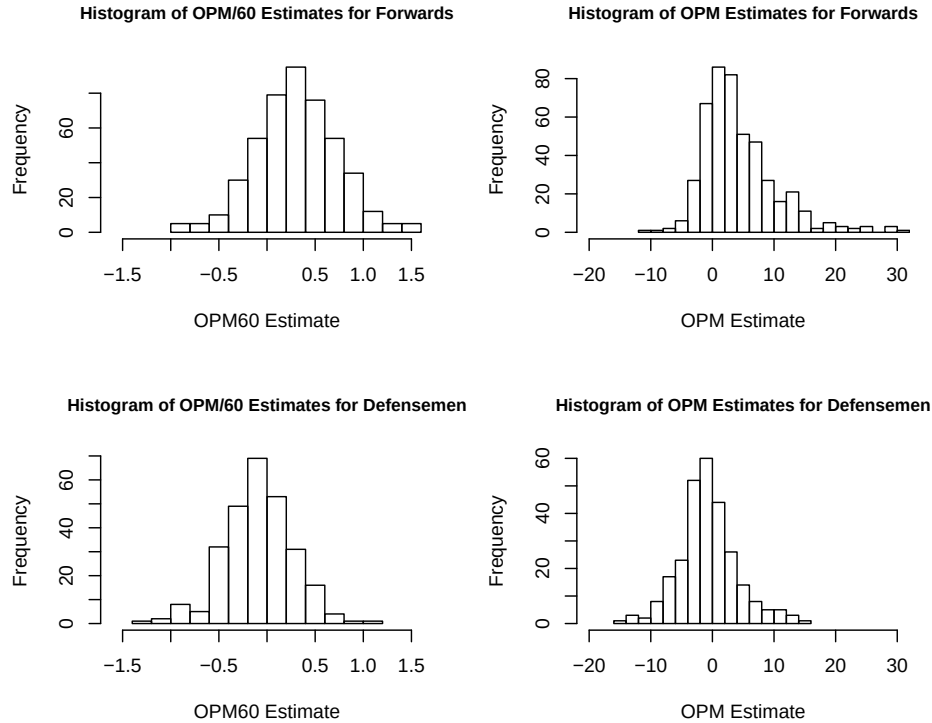
Table 3.1.1: Top 10 Skaters in OPM60

Rk	Player	Pos	OPM60	DPM60	APM60	AErr60	GF60	Mins
1	Sidney Crosby	C	1.583	-0.173	1.410	0.583	3.61	1063
2	Henrik Sedin	C	1.507	-0.626	0.880	0.924	3.35	1178
3	Pavel Datsyuk	C	1.500	0.573	2.073	0.691	3.37	1200
4	Marian Gaborik	RW	1.496	0.530	2.025	0.615	3.31	865
5	Alexander Radulov	RW	1.481	-0.359	1.122	0.869	3.62	348
6	Alex Ovechkin	LW	1.400	0.005	1.404	0.715	3.69	1272
7	Evgeni Malkin	C	1.291	-0.276	1.014	0.559	3.36	1172
8	Jakub Voracek	RW	1.267	-0.011	1.255	0.736	3.05	623
9	Eric Staal	C	1.235	0.025	1.259	0.622	2.98	1168
10	Zach Parise	LW	1.228	0.194	1.422	0.656	2.91	1176

many as being among the best offensive players in the game, with the exception of a two players with low minutes played and high errors: Alexander Radulov and Jakub Voracek. Those players have far fewer minutes than the other players, and their estimates are less reliable. Interestingly, Henrik Sedin actually has the highest standard error in the list, even higher than Radulov and Voracek, despite the fact that he has much higher minutes played totals. This is likely due to the fact that he spends most of his time playing with his brother Daniel (see Section 4.6).

Forwards dominated the list in Table 3.1.1. That forwards are more prevalent than defensemen on this list is not unexpected, as one would probably assume that forwards contribute to offense more than defensemen do. We can see this trend more clearly by plotting a histogram for $OPM/60$ results for both forwards and defensemen. See Figure 3.1.1. The histogram for forwards lies to the right of the histogram for defensemen, suggesting that the $OPM/60$ estimates for forwards are generally higher than the $OPM/60$ estimates for defensemen.

Since forwards dominated the list of top players in $OPM/60$, we give the top defensemen in $OPM/60$ in Table 3.1.2. There are some top offensive defensemen in this list, along with some players with low minutes, high errors, and skeptical ratings. One example is Carl Gunnarsson, who tops this list and is one of the players with low minutes. Gunnarsson did have a decent season offensively in 2009-10, scoring 12 even-strength points in 43 games, while playing just the 6th most minutes per game among defensemen on his team. Projecting his statistics over 82 games, Gunnarsson would have had 23 even-strength points, tying him with Tomas Kaberle for the team lead among defensemen, despite playing less minutes. So we do get an idea of why the model gave him this high estimate. We note that the lower

Figure 3.1.1: Kernel Density Estimation for $OPM/60$ Estimates and OPM Estimates.Table 3.1.2: Top 10 Defensemen in $OPM60$

Rk	Player	Pos	OPM60	DPM60	APM60	AErr60	GF60	Mins
1	Carl Gunnarsson	D	1.177	0.537	1.714	0.978	2.94	238
2	Ville Koistinen	D	0.911	0.409	1.320	0.840	2.94	381
3	Cody Franson	D	0.789	0.807	1.596	1.039	2.95	244
4	Mike Green	D	0.713	0.223	0.935	0.588	3.31	1347
5	Mark Streit	D	0.687	-0.287	0.400	0.541	2.69	1212
6	Andrei Markov	D	0.664	-0.047	0.617	0.619	2.69	1117
7	Stephane Robidas	D	0.580	0.324	0.904	0.586	2.60	1323
8	Johnny Oduya	D	0.578	0.148	0.726	0.569	2.78	1223
9	Bret Hedican	D	0.561	-0.392	0.169	0.660	2.97	600
10	Jaroslav Spacek	D	0.518	-0.074	0.444	0.565	2.86	1189

end of the 95% confidence interval for Gunnarsson's $OPM/60$ is near zero, suggesting that, at worst, he was still an average offensive defenseman at even-strength during the limited minutes that he played.

3.2 OPM

Recall that OPM is a measure of the offensive contribution of a player at even-strength in terms of goals over an entire season. The top 10 players in OPM were already given and discussed in the introduction, Table 1.1.1. That list was dominated by forwards, a trend that can also be seen in Figure 3.1.1, so we now discuss the top 10 defensemen in OPM given in Table 3.2.1. Most of the players in Table

Table 3.2.1: Top 10 Defensemen in OPM

Rk	Player	Pos	OPM	DPM	APM	AErr	GF	Mins
1	Mike Green	D	16.0	5.0	21.0	13.2	74	1347
2	Mark Streit	D	13.9	-5.8	8.1	10.9	54	1212
3	Stephane Robidas	D	12.8	7.2	19.9	12.9	57	1323
4	Andrei Markov	D	12.4	-0.9	11.5	11.5	50	1117
5	Johnny Oduya	D	11.8	3.0	14.8	11.6	57	1223
6	Ian White	D	11.6	1.1	12.7	11.3	61	1350
7	Duncan Keith	D	10.8	12.7	23.4	16.9	76	1546
8	Dion Phaneuf	D	10.7	-1.5	9.2	12.6	67	1448
9	Jaroslav Spacek	D	10.3	-1.5	8.8	11.2	57	1189
10	Dan Boyle	D	10.0	-5.6	4.4	11.0	52	1180

3.2.1 are typically considered among the top offensive defensemen in the league at even-strength. Nicklas Lidstrom is one notable omission. Lidstrom is 15th among defensemen with an OPM of 7.7. Despite the seemingly low offensive rating, Lidstrom will rise to the top of the APM list below, thanks to a league best DPM rating.

3.3 $DPM/60$

Recall that $DPM/60$ is a measure of the defensive contribution of a player in terms of goals per 60 minutes of playing time at even-strength. Without specifying a minimum minutes played limit, we get several players with low minutes in the list of top players in $DPM/60$ given in Table 3.3.1. In order to remove the players with low minutes from this list, we restrict the list to those players with more than 700

Table 3.3.1: Top 10 Skaters in DPM60 (minimum 0 minutes)

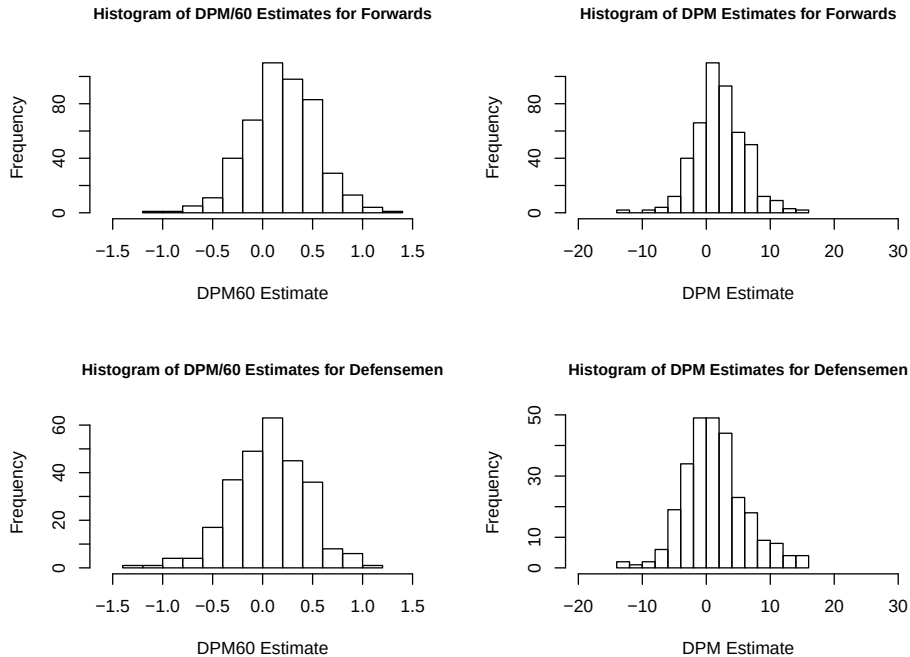
Rk	Player	Pos	OPM60	DPM60	APM60	AErr60	GA60	Mins
1	Derek Dorsett	RW	0.086	1.228	1.313	0.884	1.44	320
2	Drew Miller	LW	-0.043	1.137	1.093	0.789	1.59	378
3	Peter Regin	C	0.557	1.053	1.610	0.908	1.74	322
4	Kurt Sauer	D	-0.567	1.039	0.472	0.688	1.98	656
5	George Parros	RW	0.105	1.030	1.134	0.856	1.12	393
6	Martin Gelinas	LW	0.059	1.009	1.068	1.055	1.64	219
7	Adam Hall	RW	-0.651	0.980	0.329	0.829	1.68	332
8	Ryan Callahan	RW	0.113	0.973	1.086	0.602	1.77	882
9	Paul Martin	D	-0.219	0.968	0.748	0.674	1.59	929
10	Josef Vasicek	C	0.556	0.963	1.519	0.964	1.69	343

Table 3.3.2: Top 10 Skaters in DPM60 (minimum 700 minutes)

Rk	Player	Pos	OPM60	DPM60	APM60	AErr60	GA60	Mins
1	Ryan Callahan	RW	0.113	0.973	1.086	0.602	1.77	882
2	Paul Martin	D	-0.219	0.968	0.748	0.674	1.59	929
3	Marco Sturm	LW	0.308	0.961	1.269	0.697	1.58	710
4	Jason Pominville	RW	0.648	0.863	1.511	0.690	2.29	1059
5	Mike Weaver	D	-0.944	0.819	-0.125	0.600	1.72	824
6	Willie Mitchell	D	-0.296	0.775	0.480	0.558	1.93	1193
7	Mikko Koivu	C	0.733	0.757	1.490	0.670	2.06	1041
8	David Krejci	C	0.540	0.753	1.293	0.727	1.87	929
9	Daniel Sedin	LW	0.749	0.745	1.494	0.915	2.06	1066
10	Tyler Kennedy	C	0.626	0.741	1.366	0.646	1.76	727

minutes played. The new list is given in Table 3.3.2. In that list we get a mix of forwards and defensemen. To see if this trend continues outside the top 10, we can again plot a histogram of $DPM/60$ estimates for forwards and defensemen. See Figure 3.3.1. Forwards and defensemen seem to have a fairly similar distribution of $DPM/60$ estimates, though defensemen may be slightly behind forwards. It may seem counterintuitive that defensemen have lower ratings than forwards. The estimates seem to indicate that forwards contribute more to defense (per 60 minutes of ice time) than defensemen do, but the difference in estimates is so small that it

Figure 3.3.1: Histogram of $DPM/60$ Estimates and DPM Estimates



could be simply due to noise. The trends are slightly different with DPM , which is playing-time dependent. Top defensemen typically play more minutes than top forwards, so their ratings are helped when playing-time is considered as well.

Ryan Callahan, Paul Martin, and Marco Sturm lead this list by a sizeable margin, partly because of very low $GA/60$ of 1.77, 1.59, and 1.58, respectively. Sturm has just above the minimum for minutes played, but given his very low $GA/60$, his estimate is believable.

We now discuss the top 10 in $DPM/60$ for forwards and defensemen separately. In the list of top forwards in $DPM/60$ given in Table 3.3.3, we get some well-known defensive forwards, and a couple interesting names. Once again a Sedin twin has the highest errors in a list. The model seems to be giving Daniel the credit for the Sedin line's success in $DPM/60$, whereas for $OPM/60$, Henrik gets the credit. Henrik's $DPM/60$ estimate is actually negative (-0.626), while his $OPM/60$ estimate (1.506) is much higher than Daniel's $OPM/60$ estimate (0.749).

Tyler Kennedy is part of what many hockey analysts consider to be one of the top defensive lines in hockey, and his $DPM/60$ estimate supports that belief. We note that Jordan Staal's $DPM/60$ estimate is 0.412 , and he has a higher $GA/60$ (1.95) than Kennedy (1.76), so we can see one possible reason why the model gave

Table 3.3.3: Top 10 Forwards in DPM60 (minimum 700 minutes)

Rk	Player	Pos	OPM60	DPM60	APM60	AErr60	GA60	Mins
1	Ryan Callahan	RW	0.113	0.973	1.086	0.602	1.77	882
2	Marco Sturm	LW	0.308	0.961	1.269	0.697	1.58	710
3	Jason Pominville	RW	0.648	0.863	1.511	0.690	2.29	1059
4	Mikko Koivu	C	0.733	0.757	1.490	0.670	2.06	1041
5	David Krejci	C	0.540	0.753	1.293	0.727	1.87	929
6	Daniel Sedin	LW	0.749	0.745	1.494	0.915	2.06	1066
7	Tyler Kennedy	C	0.626	0.741	1.366	0.646	1.76	727
8	Tomas Plekanec	C	0.051	0.727	0.778	0.663	2.15	1058
9	Colby Armstrong	RW	0.470	0.697	1.167	0.581	2.37	947
10	Manny Malhotra	C	0.309	0.647	0.955	0.555	1.84	933

Table 3.3.4: Top 10 Defensemen in DPM60 (minimum 700 minutes)

Rk	Player	Pos	OPM60	DPM60	APM60	AErr60	GA60	Mins
1	Paul Martin	D	-0.219	0.968	0.748	0.674	1.59	929
2	Mike Weaver	D	-0.944	0.819	-0.125	0.600	1.72	824
3	Willie Mitchell	D	-0.296	0.775	0.480	0.558	1.93	1193
4	Andrew Greene	D	-0.427	0.723	0.296	0.618	1.68	1013
5	Jan Hejda	D	0.088	0.714	0.803	0.611	2.24	1285
6	Nicklas Lidstrom	D	0.333	0.677	1.010	0.708	1.84	1380
7	Ken Klee	D	0.018	0.646	0.663	0.670	2.27	722
8	Sean O'Donnell	D	0.012	0.596	0.608	0.523	1.76	1195
9	Rob Scuderi	D	-0.125	0.581	0.457	0.534	1.95	1157
10	Radek Martinek	D	-0.382	0.568	0.185	0.736	2.35	800

Kennedy the higher estimate of the two linemates. Incidentally, if we weight the observations in the model so that the 2009-2010 season counts more heavily than the other two seasons, Staal makes the list of top 10 forwards in $DPM/60$. This estimate supports his nomination for the 2009-2010 Selke trophy, which is given each season to the top defensive forward in the game.

Mike Weaver is second in the list of top defensemen in $DPM/60$, which is shown in Table 3.3.4. Weaver has the third lowest $GA/60$ in this list at 1.73, and his most common teammates, Carlo Colaiacovo (2.40) and Brad Boyes (2.638) have a higher $GA/60$, so we see one reason why the model gave him a high $DPM/60$ rating. Unfortunately, Weaver has the worst $OPM/60$ in the league (minimum 700

minutes). In fact, many of the players on this list have low offensive ratings, and Nicklas Lidstrom (0.333) has the only high rating.

3.4 DPM

Recall that *DPM* is a measure of the defensive contribution of a player at even-strength in terms of goals over an entire season. We now discuss the top 10 skaters, forwards, and defensemen in *DPM*.

Table 3.4.1: Top 10 Skaters in DPM

Rk	Player	Pos	OPM	DPM	APM	AErr	GA	Mins
1	Nicklas Lidstrom	D	7.7	15.6	23.2	16.3	42	1380
2	Willie Mitchell	D	-5.9	15.4	9.5	11.1	38	1193
3	Jan Hejda	D	1.9	15.3	17.2	13.1	48	1285
4	Jason Pominville	RW	11.4	15.2	26.7	12.2	40	1059
5	Paul Martin	D	-3.4	15.0	11.6	10.4	25	929
6	Ryan Callahan	RW	1.7	14.3	16.0	8.8	26	882
7	Jay Bouwmeester	D	-9.7	13.9	4.2	13.0	57	1551
8	Daniel Sedin	LW	13.3	13.2	26.5	16.3	37	1066
9	Mikko Koivu	C	12.7	13.1	25.8	11.6	36	1041
10	Tomas Plekanec	C	0.9	12.8	13.7	11.7	38	1058

Unlike the list of top skaters in *DPM*/60 (Table 3.3.2), defensemen are more prevalent than forwards on the list of top skaters in *DPM* given in Table 3.4.1, due to their higher minutes played. Defensemen make up 5 of the first 7, and 10 of the first 15 skaters in *DPM*. Beyond the top 15, the distribution of *DPM* estimates for forwards are actually very similar (see Figure 3.3.1).

We now look at forwards and defensemen separately. Many of the top forwards in Table 3.4.2 are known to be very solid defensive forwards. Ryan Callahan made the United States Olympic team in part because he is very good defensively, Mikko Koivu is often praised for his excellent two-way play, and Pavel Datsyuk is a two-time Selke Trophy winner for the best defensive forward in the league. Jason Pominville's ranking is surprising given that his *GA*/60 (2.29) is the worst among players on this list. Checking his most common linemates, we find Jochen Hecht (2.64), Toni Lydman (2.36), and Tim Connolly (2.305) whose *GA*/60 are not significantly different than Pominville's. Pominville did lead his team in traditional plus-minus in 2007-2008 (+16), and was tied for second in 2009-2010 (+13), which could be one reason for the high rating, but he was also a -4 in 2008-2009.

Table 3.4.2: Top 10 Forwards in DPM

Rk	Player	Pos	OPM	DPM	APM	AErr	GA	Mins
1	Jason Pominville	RW	11.4	15.2	26.7	12.2	40	1059
2	Ryan Callahan	RW	1.7	14.3	16.0	8.8	26	882
3	Daniel Sedin	LW	13.3	13.2	26.5	16.3	37	1066
4	Mikko Koivu	C	12.7	13.1	25.8	11.6	36	1041
5	Tomas Plekanec	C	0.9	12.8	13.7	11.7	38	1058
6	David Krejci	C	8.4	11.7	20.0	11.3	29	929
7	Pavel Datsyuk	C	30.0	11.5	41.5	13.8	38	1200
8	Marco Sturm	LW	3.6	11.4	15.0	8.2	19	710
9	Derek Roy	C	11.4	11.2	22.5	12.4	48	1122
10	Colby Armstrong	RW	7.4	11.0	18.4	9.2	37	947

Table 3.4.3: Top 10 Defensemen in DPM

Rk	Player	Pos	OPM	DPM	APM	AErr	GA	Mins
1	Nicklas Lidstrom	D	7.7	15.6	23.2	16.3	42	1380
2	Willie Mitchell	D	-5.9	15.4	9.5	11.1	38	1193
3	Jan Hejda	D	1.9	15.3	17.2	13.1	48	1285
4	Paul Martin	D	-3.4	15.0	11.6	10.4	25	929
5	Jay Bouwmeester	D	-9.7	13.9	4.2	13.0	57	1551
6	Duncan Keith	D	10.8	12.7	23.4	16.9	56	1546
7	Tobias Enstrom	D	-1.6	12.4	10.8	13.9	59	1329
8	Andrew Greene	D	-7.2	12.2	5.0	10.4	28	1013
9	Sean O'Donnell	D	0.2	11.9	12.1	10.4	35	1195
10	Ryan Suter	D	2.4	11.8	14.2	16.2	51	1360

Paul Martin, the leader among defensemen in $DPM/60$, is fourth on the list of best defensemen in DPM given in Table 3.4.3, despite much lower minutes played than the others. Martin has a nice list of most common teammates (Martin Brodeur, 1.92 $GA/60$; Johnny Oduya, 2.00; Zach Parise, 1.74) but his $GA/60$ (1.59) is extremely low, which is probably the cause of his low $DPM/60$ and DPM estimates.

Tobias Enstrom's *DPM*/60 and *DPM* estimates are high given that his 2.68 *GA*/60 and 59 *GA* statistics are the worst in the list. However, Enstrom's *GA*/60 is better than the *GA*/60 of his most common linemates, Niclas Havelid (2.84 *GA*/60), Ilya Kovalchuk (3.07), and Todd White (3.06). Another Atlanta defenseman, Ron Hainsey, also has a high *DPM* given his raw statistics. Looking deeper, his 2.50 *GA*/60 is actually second to only Colby Amrstrong (2.37) among players with more than 700 minutes on his team. Our model seems to be saying that Hainsey, like Enstrom, is better than his raw statistics suggest, mostly because of the quality of teammates that he plays with.

3.5 *APM*/60

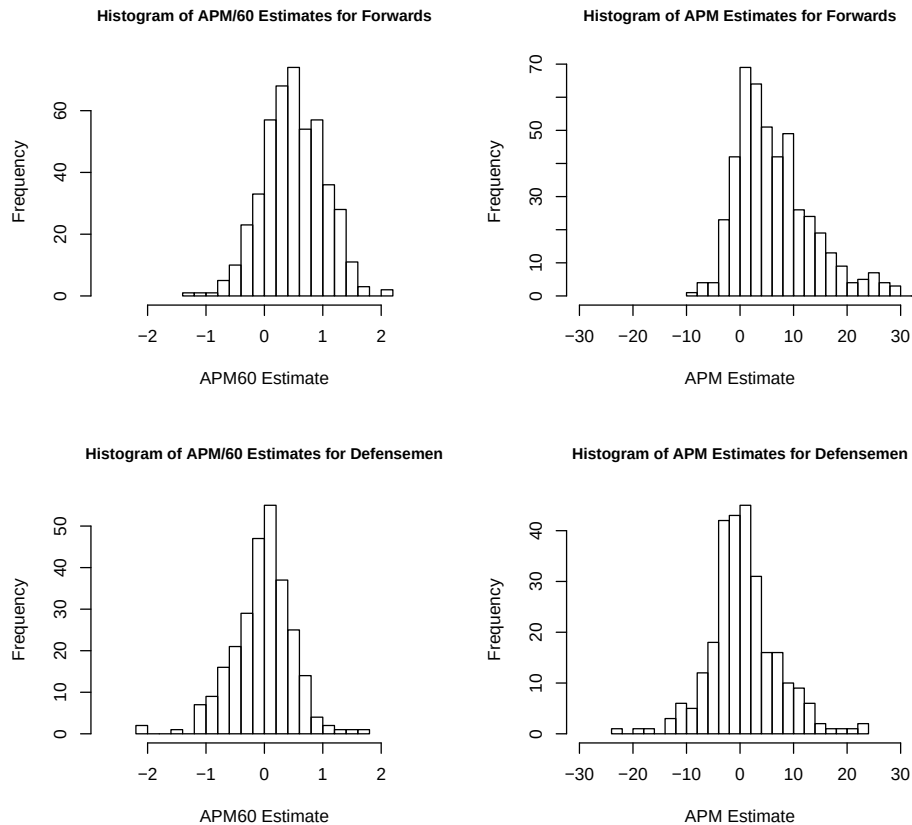
We now begin to look at the top players in the league in terms of *APM*/60 and *APM*. Recall that *APM*/60 is a measure of the total (offensive and defensive) contribution of a player at even-strength in terms of net goals (goals for minus goals against) per 60 minutes of playing time.

Table 3.5.1: Top 10 Skaters in *APM*60 (minimum 700 minutes)

Rk	Player	Pos	OPM60	DPM60	APM60	AErr60	NG60	Mins
1	Pavel Datsyuk	C	1.500	0.573	2.073	0.691	1.48	1200
2	Marian Gaborik	RW	1.496	0.530	2.025	0.615	1.04	865
3	Tim Connolly	C	0.993	0.531	1.525	0.684	0.96	711
4	Jason Pominville	RW	0.648	0.863	1.511	0.690	0.58	1059
5	Alex Burrows	LW	0.923	0.577	1.500	0.588	0.91	1033
6	Daniel Sedin	LW	0.749	0.745	1.494	0.915	1.22	1066
7	Mikko Koivu	C	0.733	0.757	1.490	0.670	0.52	1041
8	Marc Savard	C	0.873	0.598	1.471	0.712	0.71	932
9	Zach Parise	LW	1.228	0.194	1.422	0.656	1.14	1176
10	Sidney Crosby	C	1.583	-0.173	1.410	0.583	1.03	1063

Datsyuk, considered by many to be the best two-way player in the game, tops the list of best players in *APM*/60 given in Table 3.5.1. Forwards dominate this list. We plot the histogram for *APM*/60 and *APM* in Figure 3.5.1 to get an idea of whether this trend continues outside the top 10. Forwards seem to have higher estimates than defensemen. Note that the picture changes slightly for *APM*, but defensemen still seem to have lower estimates in general.

Burrows' case is an interesting one. He did not appear in the top 10 lists for *OPM*/60 or *DPM*/60, but makes the top *APM*/60 list for skaters in Table 3.5.1 with

Figure 3.5.1: Histogram of $APM/60$ Estimates and APM Estimates.

balanced offensive and defensive estimates. He played frequently with the Sedin twins in 2009-2010, so one might think his rating would be difficult to separate from the twins' estimates. However, on average over the last three years, the Sedins are not among Burrows' three most frequent linemates, so his high estimates can not be attributed to statistical noise caused by frequently playing with the twins. Burrows actually has the second lowest errors in this list of players, probably because of the varied linemates that he has had over the past three years.

Huskins and Schultz sneak onto the list of top defensemen in $APM/60$ in Table 3.5.2 with balanced ratings, despite being held off of the $OPM/60$ and $DPM/60$ top 10 lists. We note that Schultz's most common linemates are Mike Green, Alexander Ovechkin, Nicklas Backstrom, and Alexander Semin. Schultz has accumulated fairly low goal and assist totals over the past three seasons while playing with some of league's best offensive players, and yet his $OPM/60$ estimates are still high. In his case, the model may not be properly separating his offensive contribu-

Table 3.5.2: Top 10 Defensemen in APM60 (minimum 700 minutes)

Rk	Player	Pos	OPM60	DPM60	APM60	AErr60	NG60	Mins
1	Nicklas Lidstrom	D	0.333	0.677	1.010	0.708	1.07	1380
2	Mike Green	D	0.713	0.223	0.935	0.588	1.01	1347
3	Duncan Keith	D	0.418	0.491	0.910	0.655	0.75	1546
4	Stephane Robidas	D	0.580	0.324	0.904	0.586	0.14	1323
5	Jan Hejda	D	0.088	0.714	0.803	0.611	0.06	1285
6	Paul Martin	D	-0.219	0.968	0.748	0.674	0.86	929
7	Johnny Oduya	D	0.578	0.148	0.726	0.569	0.77	1223
8	Niklas Kronwall	D	0.358	0.360	0.717	0.684	0.63	1072
9	Kent Huskins	D	0.426	0.279	0.705	0.602	0.73	928
10	Jeff Schultz	D	0.460	0.208	0.668	0.633	1.11	1102

tion from those of his teammates. It is also possible that Schultz does a lot of little things on the ice that do not appear in box score statistics, but that contribute to his team's offensive success nonetheless. In hockey, it is difficult to separate offense and defense. A good defensive team, which can clear the puck from the defensive zone quickly, can help its offense by increasing its time of possession. Likewise, a team with a good puck possession offense can help its defense by simply keeping the puck away from the opposition. See Pronman (2010a) and Pronman (2010b) for a discussion by Corey Pronman about the connection between offense and defense.

3.6 APM

Recall that *APM* is a measure of the total (offensive and defensive) contribution of a player at even-strength in terms of net goals over an entire season. A hockey fan familiar with the traditional plus-minus statistic can think of *APM* in the same way, remembering that *APM* has been adjusted for both the strength of a player's teammates and the strength of his opponents.

Pavel Datysuk has won the Selke Trophy in the 2007-08 and 2008-09 seasons, and is widely regarded as one of the top two-way players in the game, at least among forwards. He leads the list of top skaters in Table 3.6.1 by a wide margin. It should also be pointed out that all of the other players on this list are still within two standard errors of the top spot. Interestingly, Alex Ovechkin has the lowest defensive estimates on this list, which hurts his overall ratings. Sidney Crosby's defensive rating lowered his overall rating as well, but the main reason he does not appear on this list is that he missed 29 games due to injury in the 2007-2008 season.

Table 3.6.1: Top 10 Skaters in APM

Rk	Player	Pos	OPM	DPM	APM	AErr	NG	Mins
1	Pavel Datsyuk	C	30.0	11.5	41.5	13.8	29	1200
2	Alex Ovechkin	LW	29.7	0.1	29.8	15.2	30	1272
3	Marian Gaborik	RW	21.6	7.6	29.2	8.9	15	865
4	Joe Thornton	C	23.0	5.6	28.5	12.6	22	1228
5	Zach Parise	LW	24.1	3.8	27.9	12.9	22	1176
6	Jason Pominville	RW	11.4	15.2	26.7	12.2	11	1059
7	Daniel Sedin	LW	13.3	13.2	26.5	16.3	21	1066
8	Martin St. Louis	RW	19.7	6.5	26.1	12.9	1	1297
9	Mikko Koivu	C	12.7	13.1	25.8	11.6	9	1041
10	Alex Burrows	LW	15.9	9.9	25.8	10.1	16	1033

Table 3.6.2: Top 10 Defensemen in APM

Rk	Player	Pos	OPM	DPM	APM	AErr	NG	Mins
1	Duncan Keith	D	10.8	12.7	23.4	16.9	20	1546
2	Nicklas Lidstrom	D	7.7	15.6	23.2	16.3	25	1380
3	Mike Green	D	16.0	5.0	21.0	13.2	22	1347
4	Stephane Robidas	D	12.8	7.2	19.9	12.9	3	1323
5	Jan Hejda	D	1.9	15.3	17.2	13.1	1	1285
6	Johnny Oduya	D	11.8	3.0	14.8	11.6	16	1223
7	Ryan Suter	D	2.4	11.8	14.2	16.2	4	1360
8	Fedor Tyutin	D	3.5	10.0	13.5	12.5	2	1334
9	Zdeno Chara	D	9.2	3.8	13.0	15.1	17	1455
10	Niklas Kronwall	D	6.4	6.4	12.8	12.2	12	1072

Since forwards dominated the list in Table 3.6.1, we list the top 10 defensemen separately in Table 3.6.2. The top 3 in *APM*/60 (Table 3.5.2) are once again in the top 3 here, though in a different order. One player we have not discussed is Jan Hejda, who seems to be an underrated player according to his *APM* estimate. Looking at his traditional box score statistics, we see that during both the 2007-2008 season (+20, 13 more than the second highest Blue Jacket) and the 2008-2009 season (+23, 11 more than the second highest Blue Jacket) seasons, Hejda led his team in plus-minus by a wide margin. His numbers were much worse during his

injury shortened 2009-2010 season in which the entire Blue Jackets team struggled defensively, but his performance in the previous two seasons is one reason the model could be giving him a high estimate for *DPM*. Also, his two most common linemates, Mike Commodore (2.73) and Rick Nash (2.68) have a higher *GA/60* than does Hejda (2.24), which may be helping his defensive rating.

4 Discussion of the Model

We now discuss several aspects of the formation and analysis of our model. In Section 4.1 we summarize some advantages of *APM*, including that *APM* is independent of teammates, opponents, and box score statistics. We discuss disadvantages of *APM* in Section 4.2, including statistical noise and difficulties in computing the estimates. We discuss the selection of the explanatory variables and response variables in Section 4.3, and selection of the observations in Section 4.4. Closely related to the selection of the components in the model are the assumptions that we made, and we discuss those in Section 4.5. The main assumptions discussed in that section are the exclusion of goalies in the model, and that there are no interactions between players. One of the main disadvantages of *APM* listed in Section 4.2 is the errors associated with the estimates, and we discuss these errors in greater detail in Section 4.6. Also in that section, we give top 10 lists of the players with the highest and lowest errors in *APM/60* and *APM*. Finally, in Section 4.7, we finish with some concluding remarks and give two ideas for future work: modeling special teams situations, and accounting for the zone in which each shift starts.

4.1 Advantages of *APM*

As with any metric of its kind, *APM* has its advantages and disadvantages. As we have mentioned previously, the most important benefit of *APM* is that a player's *APM* does not depend on the strength of that player's teammates or opponents. A major downside of the traditional plus-minus statistic is that it *does* depend on both teammates and opponents, so it is not always a good measure of a player's individual contribution. For example, a player on a below-average team could have a traditional plus-minus that is lower than average simply because of the linemates he plays with on a regular basis. An average player on a hypothetical line with Wayne Gretzky and Mario Lemieux would probably have a traditional plus-minus that is very high, but that statistic would not necessarily be a good measure of his contribution to his team. On the other hand, the coefficients in our model, which we use to estimate a player's *APM*, are a measure of the contribution of a player when he is on the ice, independent of all other players on the ice.

Another benefit of *APM* is that the estimate, in theory, incorporates all aspects of the game, not just those areas that happen to be measured by box score statistics. Box score statistics do not describe everything that happens on the ice. For example, screening the goalie on offense and maintaining good positioning on defense are two valuable skills, but they are not directly measured using box score statistics. *APM* is like traditional plus-minus in that it attempts to measure how a player effects the outcome on the ice in terms of goals scored by his team on offense and goals allowed by his team on defense. A player's personal totals in goals, assists, points, hits, and blocked shots, for example, are never used in computing *APM*. Nothing is assumed about the value of these box score statistics and how they impact a player's and a team's performance.

Another benefit of our model is that we make minimal *ad hoc* assumptions about which positions deserve the most credit for goal scoring or goal prevention. We do not assume, for example, that defensemen deserve more credit than forwards in goal prevention, or that forwards deserve more of the credit when a goal is scored. From Figure 3.1.1, it seems that forwards contribute more than defensemen to goal scoring, but no such assumption was made during the formation of the model.

4.2 Disadvantages of *APM*

One main drawback of the *APM* estimates is statistical noise. A priority in future research is to take measures to reduce the errors. We discuss the errors in detail in Section 4.6. Another drawback of *APM* is that the estimates do not include shootouts, and do not include the value of either penalties drawn by a player or penalties taken by a player. A team's performance in shootouts has a big impact on their place in the standings. Shootout specialists can be very valuable to a team during the regular season, and ideally shootout performance would be accounted for in *APM*. Penalties drawn and taken also impact the outcome of a game. Penalties drawn by a player lead to more power plays for that player's team, which in turn leads to more goals for his team. Likewise, penalties taken by a player lead to more power plays for, and more goals for, the opposing team. If the value of shootout performance, penalties drawn, and penalties taken were estimated using another method that gives results the units of goals per game or goals per season, those values could easily be combined with *APM*.

Another difficulty with *APM* is that the data required to calculate it is large, difficult to obtain in a usable form, and difficult to work with. Collecting and managing the data was easily the most time-consuming aspect of this research. Also, the data required for this model is only available (at least publicly) for very recent

seasons. This model could not be used to estimate the value of Wayne Gretzky, Mario Lemieux, and Bobby Orr, for example, independent of their teammates and opponents.

The final downside to *APM* is that the model requires knowledge of linear regression or linear algebra, and is not easily computed from traditional statistics. The mathematics required makes the calculation of *APM* accessible to fewer hockey fans. It was a priority to ensure that at least the estimates themselves could be easily understood, even if the methods of calculating them are not.

4.3 Selection of the variables

We now make a few remarks on how we chose the explanatory variables and the response variable in the model. For the model, we included players who played more than 3000 shifts during the 2007-2008, 2008-2009, and 2009-2010 seasons. In terms of minutes played, this cutoff is roughly 200 minutes per season on average. Players with less than 3000 shifts during those seasons would have very noisy estimates which would not be very reliable. Increasing the 3000 shift cutoff would have reduced the errors slightly, but we would have also obtained estimates for fewer players.

In our model, the units of the coefficients are the same as the units of y . We wanted to estimate a player's contribution to his team in terms of goals per 60 minutes, so we chose our response variable with those same units. The choice of goals per 60 minutes for the units of our estimates was important because we could rate players based on this statistic, and we could also convert this rate statistic to a counting statistic, total goals over an entire season, using the minutes played by each player. The resulting estimates have the units of goals, and they can be easily compared with traditional plus-minus as well as advanced metrics already in existence. Also, a priority was to ensure our estimates could be easily interpreted by the average hockey fan. Since the units of *APM* are goals over an entire season, any hockey fan familiar with traditional plus-minus can understand the meaning of *APM*.

4.4 Selection of the observations

Recall that we define a shift to be a period of time during the game when there are no substitutions made. During the 2007-2008, 2008-2009, and 2009-2010 seasons, there were 990,861 shifts. We consider only shifts that take place at even-strength (5-on-5, 4-on-4, 3-on-3), and we also require that two goalies be on the ice. All power play and empty net situations were removed.

We noticed some errors in the data. There is a minimum of four players (counting goalies) and a maximum of 6 players (counting goalies) that can be on the ice for a team at the same time. However, for some shifts, there are less than four players or more than six players on the ice for a team. These shifts may have occurred in the middle of a line change, during which it is difficult to record in real-time which players are on or off the ice. Such shifts were removed. We also note that five games had missing data, and a few more games had incomplete data, such as data for just one or two periods. The equivalent of about 10 games of data is missing out of a total of 3,690 games during the three seasons in question. After removing shifts corresponding to empty net situations and special teams situations, and shifts where errors were identified, 798,214 shifts remained.

4.5 Discussion of assumptions

Some of the assumptions used in the model require discussion. In particular, we discuss the exclusion of goalies in the model, and the exclusion of interaction terms in the model.

4.5.1 Goalies

In our preliminary models, we included variables for goalies. We have since removed goalies for a few reasons. One reason is that the goalies $DPM/60$ estimates were noisy, perhaps due to the fact that goalies are (almost) always on the ice, and that there are very few goalies per team. Even if the goalies' $DPM/60$ estimates were not noisy, $DPM/60$ would still not be the best way to isolate and measure goalie's individual ability. Recall that the interpretation of a goalie's $DPM/60$ is goals per 60 minutes contributed by the goalie on defense, or goals per 60 minutes prevented by the goalie. We could think of $DPM/60$ as measuring the difference between a goalie's goals against average at even strength and the league's goals against average at even strength, while adjusting for the strength of the teammates and opponents of the goalie. A goalie who has a relatively low goals against average at even strength should in general have a relatively high (good) $DPM/60$ estimate. In general, this relationship was true for our preliminary results. The 10 goalies with the highest $DPM/60$ estimates also had low $GA/60$ statistics.

Unfortunately, goals against average is not the best measure of a goalie's ability. The number of goals per 60 minutes allowed by a team depends on not only the goalie's ability at stopping shots on goal, but also the frequency and quality of the shots on goal that his team allows. So goals against average is a measure not just of a goalie's ability, but also of his team's ability at preventing shots on goal.

Ideally, the model would be able to correctly determine if the goalie or the team in front of him deserves credit for a low goals against average, but that did not seem to be happening. One reason could be the relatively low number of goalies on each team. Another reason could be that there is some team-level effect not accounted for.

Unless a different way of accounting for team effects was used, the goalie ratings from this kind of model would likely not be useful. Different techniques for measuring a goalie's ability and contribution to his team would be preferred. Most methods would likely use different information, including the quality and frequency of the shots on goal that his team allows. See, for example, Ken Krzywicki's shot quality model in Krzywicki (2005) and Krzywicki (2009).

4.5.2 Interactions between players

By not including interaction terms in the model, we do not account for interactions between players. Chemistry between two particular teammates, for example, is ignored in the model. The inclusion of interaction terms could reduce the errors. The disadvantages of this type of regression would be that it is much more computationally intensive, and the results would be harder to interpret.

4.6 Discussion of errors

In the introduction, and elsewhere in this paper, we noted that Henrik and Daniel Sedin have a much higher error than other players with a similar number of shifts. One reason for this high error could be that the twin brothers spend most of their time on the ice together. Daniel spent 92% of his playing time with Henrik, the highest percentage of any other player combination where both players have played over 700 minutes. Because of this high colinearity between the twins, it is difficult to separate the individual effect that each player has on the net goals scored on the ice. It seems as though the model is giving Henrik the bulk of the credit for the offensive contributions, and Daniel most of the credit for defense. Henrik's defensive rating is strangely low given his low goals against while on the ice. Likewise, Daniel's offensive rating is unusually low.

The ten players with the highest error in $APM/60$ are shown in Table 4.6.1. Note that if we do not impose a minutes played minimum, the list is entirely made up of players who played less than 200 minutes, so we have restricted this list to players that have played more than 700 minutes on average over the last three seasons. The Sedins have significantly larger errors than the next players in the list, and all of the players in this list are ones who spent a large percent of their time on the ice with a particular teammate or two.

Table 4.6.1: Top 10 Skaters in AErr60 (minimum 700 minutes)

Rk	Player	Pos	APM60	AErr60	Mins	Teammate.1	min1
1	Henrik Sedin	C	0.880	0.924	1178	D.Sedin	83%
2	Daniel Sedin	LW	1.494	0.915	1066	H.Sedin	92%
3	Brenden Morrow	LW	0.476	0.797	814	M.Ribeiro	72%
4	Ryan Getzlaf	C	1.325	0.788	1138	C.Perry	82%
5	Corey Perry	RW	0.628	0.776	1142	R.Getzlaf	82%
6	Tomas Holmstrom	LW	0.218	0.754	737	P.Datsyuk	87%
7	Matt Bradley	RW	0.755	0.741	738	D.Steckel	51%
8	Radek Martinek	D	0.185	0.736	800	B.Witt	58%
9	David Krejci	C	1.293	0.727	929	B.Wheeler	48%
10	Jason Spezza	C	0.831	0.725	1086	D.Alfredss	59%

Table 4.6.2: Top 10 Skaters in AErr60 (minimum 700 minutes)

Rk	Player	Pos	APM60	AErr60	Mins	Teammate.1	min1
1	Jay Bouwmeester	D	0.162	0.501	1551	K.Skrastin	24%
2	Dennis Seidenberg	D	0.087	0.503	1075	E.Staal	18%
3	Ian White	D	0.565	0.504	1350	M.Stajan	29%
4	Olli Jokinen	C	0.367	0.506	1185	J.Bouwmees	25%
5	Christian Ehrhoff	D	0.537	0.507	1290	D.Murray	22%
6	Patrick O'sullivan	C	0.122	0.508	1055	A.Kopitar	30%
7	Andrej Meszaros	D	-0.168	0.511	1162	M.St. Loui	21%
8	Lee Stempniak	RW	0.903	0.512	994	I.White	20%
9	Bryan McCabe	D	0.190	0.512	1182	S.Weiss	22%
10	Ryan Whitney	D	-0.069	0.516	1134	E.Malkin	16%

The top 10 skaters in lowest standard errors are shown in Table 4.6.2. Most of these players are defensemen and have been on the ice for a high number of minutes. Every player in Table 4.6.2 has played for two or more teams during the past three seasons. Stempniak, who has the lowest minutes played on the list, probably made the list because he has played for three different teams.

4.7 Future work and conclusions

We highlight two improvements that could be made to our model. The most important addition to this work would probably be to include a player's offensive and defensive contributions in special teams situations. While performance at even-strength is a good indicator of a player's offensive and defensive value, some players seem to have much more value when they are on special teams. Teemu Selanne is an example of one player who has the reputation of being a power play goal-scoring specialist, and we could quantify his ability using an estimate that includes special teams contributions.

Another improvement we could make is accounting for whether a shift starts in the offensive zone, defensive zone, or neutral zone, and accounting for which team has possession of the puck when the shift begins. The likelihood that a goal is scored during a shift is dependent on the zone in which a shift begins and is dependent on which team has possession of the puck when the shift begins. See, for example, Thomas (2006). This fact could be affecting the estimates of some players. Players who are relied upon for their defensive abilities, for example, may start many of their shifts in their own zone. This trend could result in more goals against for those players than if they had started most of their shifts in their offensive zone. In the current model, there is no adjustment for this bias.

Unfortunately, regression diagnostics like R^2 were not reported in Ilardi and Barzilai (2008) or Rosenbaum (2004) for comparisons. Joe Sill who developed a model similar to Rosenbaum's, reported an R^2 of 0.08 for his out-of-sample prediction, suggesting that 8% of the variation in the data is explained by the model. As one might expect, due to the chaotic nature of hockey, and the lower scoring rates, the R^2 for our model was lower at 0.001, suggesting that 0.1% of the variation in y is being explained by the model. Sill reported that when he used a ridge regression, his R^2 almost doubled, and ridge regression and some other more advanced models could be tried in hockey. One could also try using weighted shots based on the work in Krzywicki (2009) instead of goals as the response variable. Including data for initial zones may also help.

We believe that APM is a useful addition to the pool of hockey metrics already in existence. The fact that APM is independent of teammates and opponents is the main benefit of the metric. APM can be improved by addressing special teams and initial zones, and reducing the statistical noise is a priority in future research. We hope that GM's, coaches, hockey analysts, and fans will find APM a useful tool in their analysis of NHL players.

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